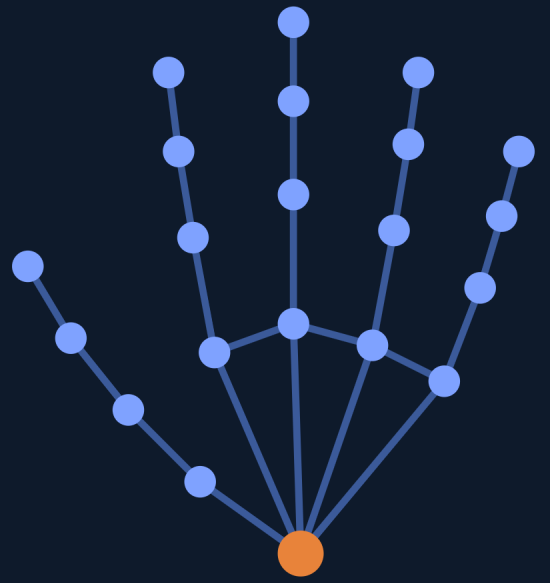
DEEP LEARNING LAB · FINAL PROJECT

SignSync

Continuous ASL-to-English translation with a part-aware graph neural network



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THE PROBLEM

Sign language carries meaning in many places at once

Hands, body and face

Handshape, movement, posture and facial expression all change the meaning.

No word boundaries

Continuous signing is one stream. The model must find where each sign starts and ends.

Not English grammar

ASL word order differs from English, so output cannot be a word-by-word mapping.

Approach: read poses, not pixels.

A skeleton is a graph of joints and bones, so a graph neural network is a natural fit.

DATASET

How2Sign: continuous ASL paired with English sentences

80+
hours of instructional ASL video

2,357
clips in the official test split

2,349
clips evaluated; 8 had no pose file

Input: 133 whole-body keypoints per frame, extracted by RTMPose.

Output: one English sentence per clip, scored against the reference.

Source: Duarte et al., How2Sign, CVPR 2021.

DATASET

How2Sign: what the data looks like

80+ h
of ASL instructional video

~35 k
sentence-level clips (approx.)

2,357
official test clips

2,349
evaluated (8 had no pose file)

Input and output

- Input: a signing clip as 133 keypoints per frame (x, y, confidence), up to 256 frames
- Output: one English sentence per clip
- Splits: train, validation and the official test set

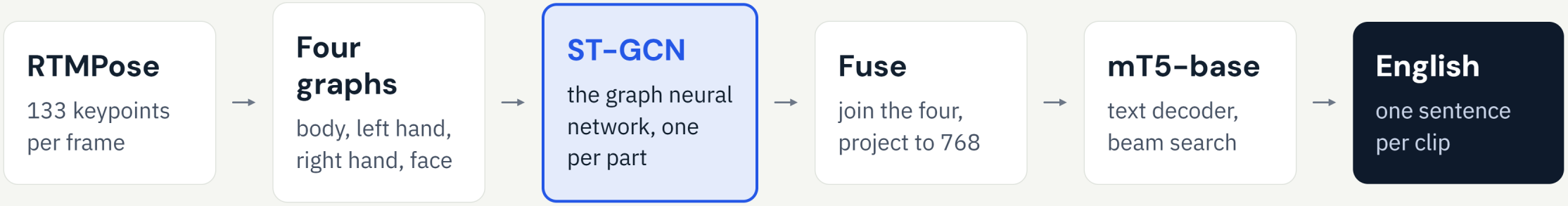
Why it is hard

- Continuous signing, with no pauses between signs
- ASL grammar and word order differ from English
- Meaning also sits in the face and body, not only the hands
- Many signers, one instructional-video domain

Source: Duarte et al., How2Sign, CVPR 2021

ARCHITECTURE

Poses in, English out: the whole pipeline



The decoder receives the prompt *“Translate sign language video to English:”* followed by the pose features, and four-beam search writes the sentence.

Blue marks the graph neural network: only 4.6 M of the model’s 588 M parameters.

THE GRAPH NEURAL NETWORK

Four skeleton graphs, one ST-GCN for each

9
Body joints

21
Left hand joints

21
Right hand joints

18
Face joints

- Spatial convolution mixes each joint with its neighbours along the bones
- Temporal convolution mixes each joint across frames
- The two hands share weights; body features feed the hand and face branches
- Mean over joints, join 4 × 256 = 1,024, project to 768 per frame

4.6 M
graph encoders

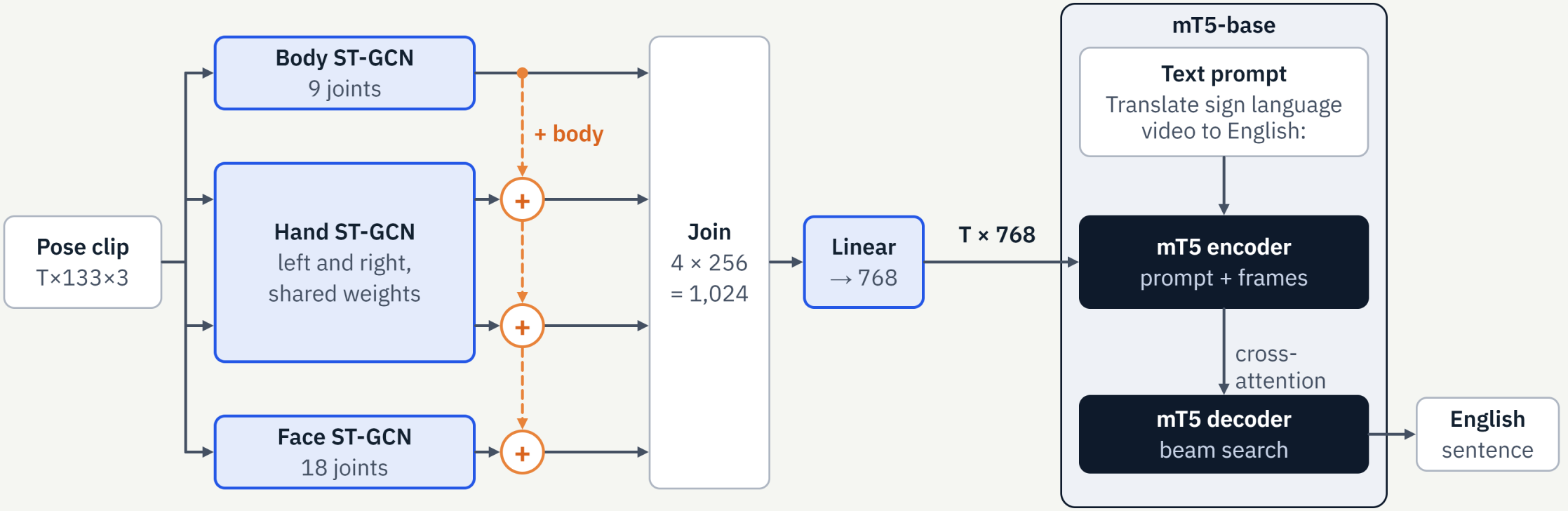
582 M
mT5-base

0.8 M
other layers

588 M
total parameters

MODEL ARCHITECTURE

Four graph encoders feed a pre-trained language model



Each part is encoded by an ST-GCN; mT5 reads the joined features after a text prompt and writes the English sentence.

REFERENCE DESIGN · THE WHOLE PROJECT

From video to a working translator, in eight stages

1
Collect
How2Sign videos paired with English sentences

2
Extract poses
RTMPose gives 133 keypoints per frame, saved as .npy

3
Prepare data
Manifest, train / val / test, four part graphs, at most 256 frames

4
Build model
Four ST-GCNs feeding mT5-base

5
Train
AdamW, cosine schedule, cross-entropy on the sentence

6
Tune
Small hyperparameter search, then decoding-only tuning

7
Evaluate
BLEU-4, ROUGE-L, bootstrap interval, control tests

8
Deploy
Web app, notebook, GitHub, hosted page

REFERENCE DESIGN · TRAINING

How the model is trained

Stage A · Pre-train

Pose-only, on large sign-video collections. Gives the graph encoder general sign knowledge.

Stage B · Fine-tune

On How2Sign for sign-to-English translation.

The mT5-base decoder starts from pre-trained weights from the four graph encoders (about 4.6 M parameters).

Optimizer	**AdamW (β 0.9, 0.999)**
Learning rate	**3e-4, warm-up then cosine decay**
Weight decay	**1e-4**
Epochs (fine-tune)	**20**
Batch size	**8, with gradient accumulation**
Input length	**up to 256 frames**

REFERENCE DESIGN · TUNING

Two kinds of tuning

Hyperparameter search

Changes the model

- Random search: 6 trials × 3 epochs on about 500 clips
- Learning rate 1e-4, 3e-4, 1e-3
- Weight decay 0, 1e-4, 1e-2
- Dropout 0, 0.1, 0.3 · batch size 4 or 8
- Keep the best validation BLEU-4, then retrain on all data

Decoding-only tuning

Changes how text is generated; no retraining

- Beams: 1, 4, 8
- Repetition penalty: 1.0, 1.2, 1.5
- No-repeat n-gram size: off or 3
- Score validation BLEU-4 and the share of looping outputs
- Pick the best; report on the test split once

RESULTS · 2,349 TEST CLIPS

13.75 BLEU-4 on 2,349 test clips

13.75

BLEU-4

95% interval 13.04 to 14.47

30.7

ROUGE-L

my own implementation

14.5

Published BLEU-4

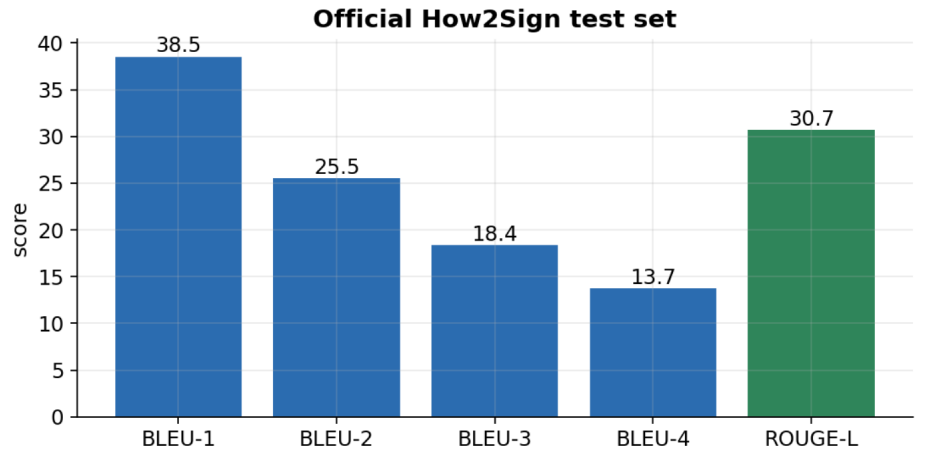
Uni-Sign paper, ROUGE 34.3

Evaluation of the released pose-only model on the official How2Sign test split. The published BLEU-4 for the same setup is 14.5.

EVALUATION · 2,349 TEST CLIPS

Evaluation scores on the How2Sign test split

Metric	Score	95% interval
BLEU-1	38.51	37.6 to 39.4
BLEU-2	25.50	24.6 to 26.4
BLEU-3	18.35	17.6 to 19.2
BLEU-4	13.75	13.1 to 14.5
ROUGE-L	30.66	29.8 to 31.5
WER	89.4%	87.6 to 91.0



Scores fall as the n-gram length grows, which is typical for translation. BLEU-4 is the headline score; WER is above 80% because a wrong word, an extra word and a missing word each count as an error.

2,349 of 2,357 test clips. Intervals from bootstrap resampling of the test sentences.

COMPARISON

SignSync against published How2Sign results

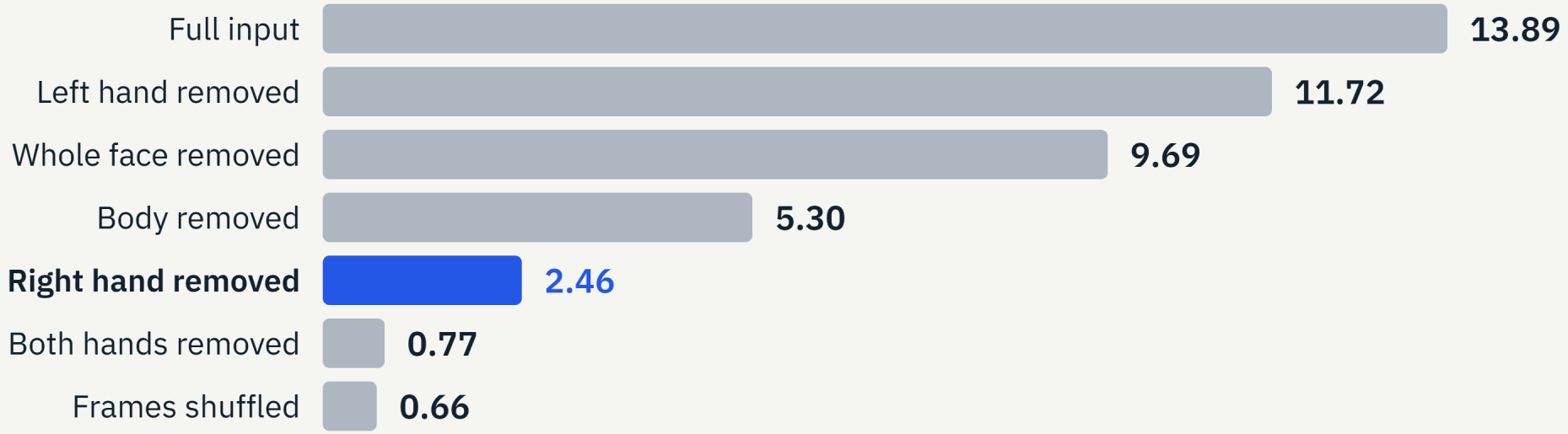
System	Setting	BLEU-4
Tarrés et al. [5]	I3D video features + Transformer	8.03
SignFormer-GCN [6]	ST-GCN keypoints + Transformer	8.53
YouTube-ASL	Large-scale pre-training, then fine-tuning	12.39
SignSync	RTMPose, four ST-GCNs, mT5-base	13.75



Published values come from each paper; check the evaluation protocol before comparing directly. Bars start at zero.

CONTROL TESTS · 500 CLIPS · BLEU-4

Removing the right hand cuts BLEU-4 from 13.89 to 2.46



Almost nothing survives without the hands or the frame order, so the score comes from the pose, not from the language model alone.

Trained model, parts zeroed at test time, no retraining. Bars start at zero.

CONTROL TESTS · 500 TEST CLIPS

Scores when one input is removed

Variant	BLEU-1	BLEU-4	ROUGE-L	WER (%)	BLEU-4 lost
Full model	39.26	13.89	30.88	88.0	0.00
Left hand zeroed	36.72	11.72	27.52	90.0	2.17
Right hand zeroed	19.62	2.46	12.68	109.2	11.43
Both hands zeroed	13.91	0.77	7.68	112.5	13.12
Body zeroed	25.43	5.30	17.24	106.0	8.59
Face zeroed	32.75	9.69	23.96	98.6	4.20
Shuffled frame order	15.15	0.66	7.09	126.8	13.23

The score collapses without the hands or without the correct frame order, so the translation is driven by the signing itself. The right hand matters most, then the body, the face and the left hand.

500 test clips. WER above 100% means the outputs were longer than the references, often because of repeated phrases.

ERROR ANALYSIS

Score by sentence length, and how outputs fail

Reference length (words)	Clips	BLEU-4
1 to 5	280	4.84
6 to 10	501	12.48
11 to 20	819	14.20
21 to 40	614	13.69
41 or more	135	7.72

Corpus BLEU-4 by reference length, 2,349 test clips.

28.7%
of outputs contain a repeated three-word phrase (674 clips)

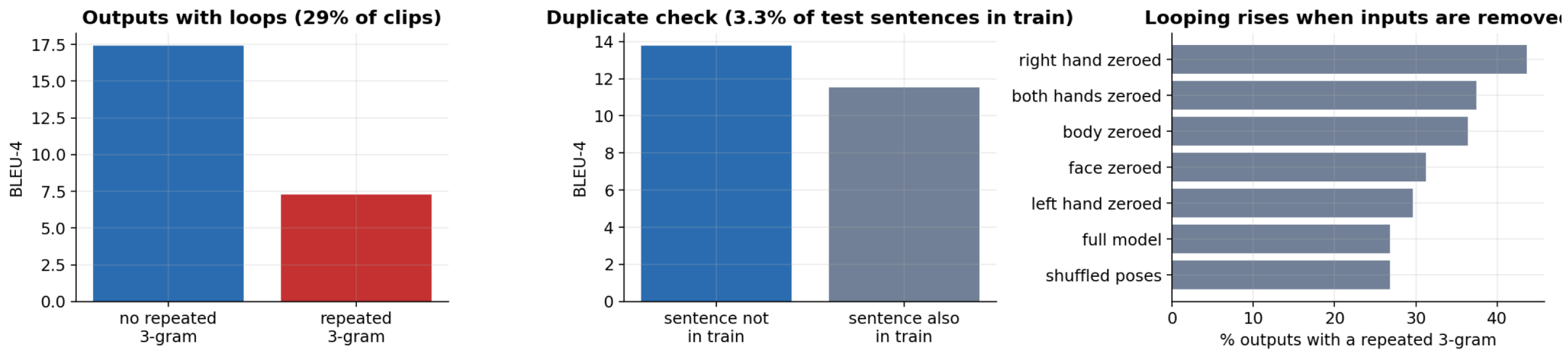
1.4%
of outputs match the reference exactly (32 clips)

97.6%
of outputs are distinct, so the model does not repeat one sentence

Medium-length sentences score best. Very short ones are scored harshly because one wrong word changes every match, and very long ones are hard to reproduce in full.

ERROR ANALYSIS

Repetition loops are the main error



- Outputs with a repetition loop score far lower: about 7 BLEU-4 against about 17 without one
- Only 3.3% of test sentences also occur in the training set, and they do not score higher, so overlap does not inflate the result
- Looping rises when informative input is removed, most of all without the right hand

DEMONSTRATION

The walkthrough, in four steps

1

Code and notebook

Every stage traced, with outputs

2

GitHub

App, scripts, results, Colab notebook

3

Deployed site

Saved results and a recorded demo

4

Live run

Upload a clip in the app and translate

github.com/Kashish811/SignSync-asl-translation

huggingface.co/spaces/Kashish811/SignSync-asl-translation

The static site shows saved results; live translation runs in the app on Colab. A recorded demo is the backup.

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QUESTIONS

Thank you

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